AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I176*

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Domain: *Healthcare,Biotechnology and Human well-being (6)*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

Rural India has no easy access to labs, so people skip blood tests and reach a doctor only when anaemia is already severe. Anaemia is highly prevalent in India, and late detection causes preventable harm

Problem Understanding

- Rural and low-income people, especially pregnant women, children and the elderly
- Labs are far and costly, technicians are scarce, and people fear needles.
- Delayed treatment, severe anaemia and overloaded district hospitals.

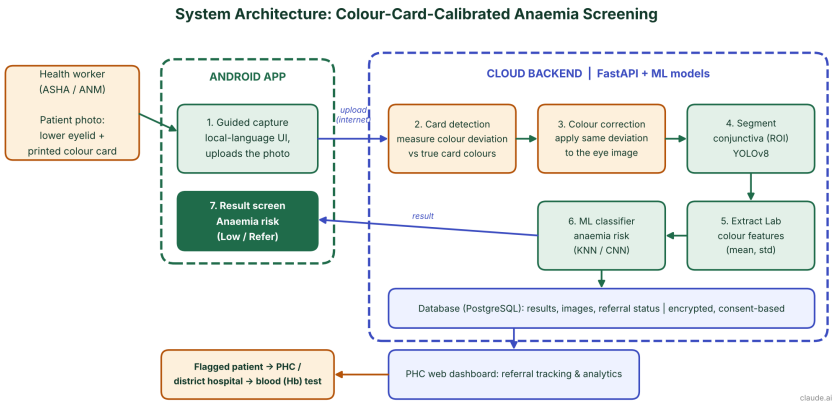
Existing Gaps & Challenges

- Lighting is not controlled.
- They depend on the internet.
- There is no low-cost way to make eye-colour readings consistent across different lighting and phones, so these apps are less reliable in real field conditions.

Our Proposed Solution

- A low-cost, offline smartphone app that screens for anaemia from a photo of the lower eyelid (conjunctiva), taken next to a printed colour card.
- A health worker photographs the patient's eye and the printed colour card together in one shot.The ML model first measures how much the card's colours have shifted from their true values because of the lighting.It applies the same correction to the eye image, so the colour is reliable.It then checks the corrected conjunctiva and gives an anaemia risk result, and people who are flagged are referred for a confirmatory blood test.
- Painless, instant, low-cost anaemia screening that works in any lighting, without a lab, needles, so people are found early instead of only when it is severe.

Technical Approach



The health worker photographs the eye with a printed colour card in the app, and the backend corrects the colours using the card. It then classifies anaemia risk from the conjunctiva and returns the result to the app. Flagged patients are tracked on the PHC dashboard for a blood test.

Tech Stack

Frontend: Flutter (Android app), React (PHC dashboard)

Backend: FastAPI (Python), hosted on cloud

Database: PostgreSQL, with cloud storage for images

AI/ML, APIs, or Hardware: OpenCV (colour-card calibration), YOLOv8 (conjunctiva segmentation), KNN / Random Forest (anaemia classification); printed colour card, Android smartphone

Innovation & Uniqueness

- A printed colour card is photographed beside the patient's eye. The system measures how far the card's colours have shifted under the lighting, then applies the same correction to the eye image before the ML analyses it.
- Existing eye-photo screening judges anaemia from colour, but sunlight, shade and different phones change that colour. Correcting the colour first makes the reading consistent in any rural setting, with no special equipment.
- Colour calibration with a low-cost printed card, which makes screening reliable without a lab, a needle or controlled lighting.

Feasibility & Scalability

- **Feasibility:** Low-cost phone, printed card and open-source software. Train on a public dataset, then validate in a small pilot against lab results.
- **Scalability:** Cloud backend serves many users, so scaling needs only more cards, apps and languages.
- **Risks:**Weak internet (queue and retry), faded card (laminated, replaceable), data diversity and false negatives (diverse data, a cautious threshold, and a blood test to confirm).

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Village health camp (ASHA worker) An ASHA worker holds a screening camp in a village with no lab. She photographs each villager's lower eyelid next to the colour card using the app. The result appears on the phone within moments, and anyone flagged is referred to the nearest PHC for a blood test. This replaces a long trip to a lab for people who feel fine.

- Use case 2: Antenatal check for a pregnant woman (ANM at a sub-centre or PHC) A pregnant woman visits for a routine antenatal check. The ANM screens her in about a minute, with no needle. If the risk is high, she is referred for a haemoglobin test and treatment, which helps prevent complications for her and the baby.

- Use case 3: Child screening at an anganwadi or school A health worker screens young children at an anganwadi centre or school. Children who are scared of needles can be checked painlessly, and flagged children are referred and tracked on the PHC dashboard until they complete the blood test.

Impacts and Benefits

- Painless, quick and free screening in their own village, with no needle, lab trip or lost wages. Pregnant women, children and the elderly are checked early, before anaemia becomes severe. ASHA and ANM workers get a simple tool that fits into visits they already do.

- Social: More people screened in places with no lab, fewer late-stage cases and fewer pregnancy and child-health complications. Economic: lower costs for families, since they avoid travel and emergency treatment. District hospitals see fewer avoidable cases, so government health spending falls over time. Environmental: fewer needles, lancets and blood-test supplies to dispose of, and fewer long trips to labs.

- Agreement of the app's result with lab haemoglobin results. Number of people screened per health worker per day. Share of flagged people who complete a confirmatory blood test. Cost per screening compared with a lab test.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [3]	Target [3.8]	Good Health and Well-being	Our tool brings low-cost anaemia screening to villages without labs, widening access to essential health services.
SDG [3]	Target [3.1]	Good Health and Well-being	Early screening of pregnant women helps catch anaemia, a risk factor in pregnancy complications, and refer them for

Individual Member Contribution

Member	Contribution or responsibility
S Dheeraaj Khaan	Team lead, Problem Statement,Problem Understanding.
R Rohit	Our Proposed Solution,Technical Approach,Innovation
Rebekah D Antony	Feasibility Scalability,Use Cases,Impacts and Benefits
S Harini	Sustainable Development Goals (SDGs),Existing Gaps and Challenges

Yukti ID

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